

1 Solution — GIAM 4.1

Problem 3

1.1 Problem

Determine the following cardinalities.

- $A = \{1, 2, \{3, 4, 5\}\}$, $|A| = \underline{\hspace{1cm}}$
- $B = \{\{1, 2, 3, 4, 5\}\}$, $|B| = \underline{\hspace{1cm}}$

1.2 Answer

$ A = 3$	$ B = 1$
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1.3 Why

Cardinality counts the elements a set has **at its top level**. It does not look inside those elements.

***Part (a).** List the elements of A one at a time, reading the outermost braces:*

$$1, \quad 2, \quad \{3, 4, 5\}.$$

That is three objects — two numbers and one set. The set $\{3, 4, 5\}$ is a single element of A , so it contributes 1 to the count, not 3. Hence $|A| = 3$.

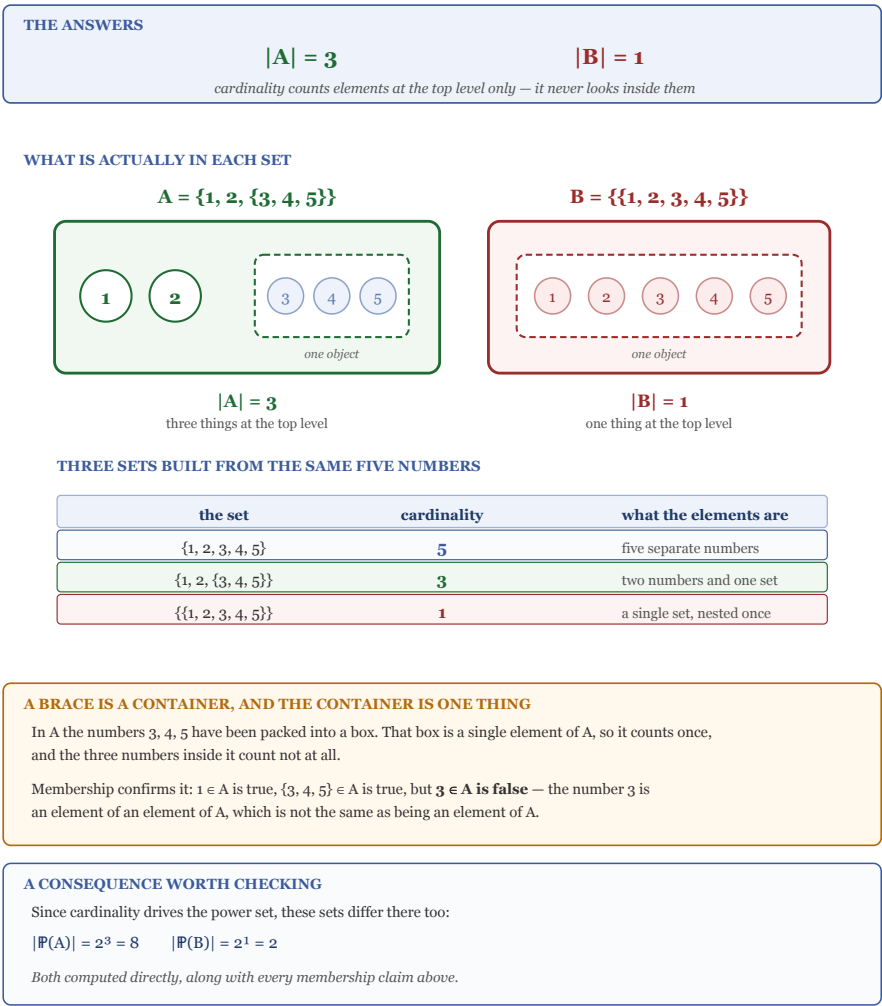
***Part (b).** The outermost braces of B enclose exactly one thing:*

$$\{1, 2, 3, 4, 5\}.$$

B is a set whose only element is a five-element set. Hence $|B| = 1$.

Both counts were confirmed directly.

1.4 The Picture



container and the container is one thing: in A the numbers 3, 4 and 5 have been packed into a box, that box is a single element of A so it counts once, and the three numbers inside it count not at all. Membership confirms it, since 1 is an element of A and the set $\{3, 4, 5\}$ is an element of A , but 3 is not an element of A — the number 3 is an element of an element of A , which is not the same as being an element of A . A final band notes a consequence worth checking: since cardinality drives the power set, the power set of A has 8 elements while the power set of B has 2, both computed directly along with every membership claim.

1.5 Remark — The Same Five Numbers, Three Different Sets

The exercise is a calibration check, and the cleanest way to see what it is checking is to line up three sets built from the same raw material:

set	cardinality	its elements
$\{1, 2, 3, 4, 5\}$	5	five numbers
$\{1, 2, \{3, 4, 5\}\}$	3	two numbers and one set
$\{\{1, 2, 3, 4, 5\}\}$	1	one set

All three “contain” the numbers 1 through 5 in a loose everyday sense, yet their cardinalities are 5, 3 and 1. What differs is only how the braces are nested — the braces are doing real work, not decoration.

A helpful image: a brace is a box. Putting three apples in a box and then putting that box on a shelf leaves **one** thing on the shelf, not three.

1.6 Remark — Membership Does Not Pass Through

The point behind part (a) is that $\in$ is not transitive. Concretely:

$$1 \in A, \quad \{3, 4, 5\} \in A, \quad \text{but} \quad 3 \notin A.$$

All three were verified. The number 3 is an element of an element of A , and that is a different relationship from being an element of A . Compare:

$$3 \in \{3, 4, 5\} \in A \quad \text{does not give} \quad 3 \in A.$$

This contrasts sharply with $\subseteq$, which **is** transitive: if $X \subseteq Y$ and $Y \subseteq Z$ then $X \subseteq Z$. Mixing the two is the standard error here, and the next section of the text is devoted to keeping them apart.

A related check: $\{1, 2\} \subseteq A$ is true, since both **1** and **2** are elements of A . But $\{3, 4, 5\} \subseteq A$ is **false** — the elements **3**, **4**, **5** are not individually in A — even though $\{3, 4, 5\} \in A$ is true. The same object satisfies one relation and fails the other.

1.7 Remark — Part (b) Is the Same Trap as $\emptyset$ Versus $\{\emptyset\}$

Part (b) is structurally identical to Problem 1 of this section. There the distinction was between

$$\emptyset \quad (0 \text{ elements}) \quad \text{and} \quad \{\emptyset\} \quad (1 \text{ element}),$$

and here it is between $\{1, 2, 3, 4, 5\}$ with five elements and $\{\{1, 2, 3, 4, 5\}\}$ with one. In both cases an extra pair of braces collapses a set of any size into a **singleton**.

That is worth stating as a rule: for any set S whatsoever,

$$|\{S\}| = 1,$$

regardless of how large S is. Wrapping something in braces always produces a one-element set. It works equally for $S = \emptyset$, for $S = \{1, 2, 3, 4, 5\}$, and for $S = \mathbb{R}$ — the singleton $\{\mathbb{R}\}$ has exactly one element.

1.8 Remark — A Consequence in the Power Set

Cardinality is not an isolated statistic; it drives everything downstream. Since $|\mathcal{P}(X)| = 2^{|X|}$,

$$|\mathcal{P}(A)| = 2^3 = 8, \quad |\mathcal{P}(B)| = 2^1 = 2,$$

both computed directly. So the difference between counting **3** and counting **5** for A is not a minor slip — it would give **32** instead of **8**, an error of a factor of four, propagating into every later calculation.

Explicitly, $\mathcal{P}(B) = \left\{ \emptyset, \left\{ \{1, 2, 3, 4, 5\} \right\} \right\}$ — the empty set and B itself, which is what one expects of a singleton's power set.